

General Overview

- TDE reads the 2 top CRPs,
- Total number of channels: 6144 (3072/CRP*2)
- Each CRP has 3072 channels for 3 views:
 - View 0 induction1 952 channels
 - View 1 induction2 952 channels
 - View 2 collection 1168 channels
- The two CRPs are read out with 10 uTCA crates containing 48 AMC cards :
 - Eight crates host 10 AMC cards, 2 crates (crate7 & crate8) host only 8 AMC cards
 - Each AMC card has 64 channels

$$(8*10*64)+(2*8*64) \\ =5120+1024=6144$$

Numbering conventions for the different readout elements

- CRP: 0,1
- View_type: 0,1,2
- View_channel: 0-951 for views 0 and 1, 0-1167 for view 2
- Crate: 0-9
- AMC: 0-9
- AMC_channel: 0-63

Two ascii files with channels mapping are provided:

1) [mapping.txt](#) (6144 lines, one line/channel)

This file has 6 columns : :

Crate; AMC; AMC_channel; CRP; view_type; view_channel

0 0 0 0 0 621
0 0 1 0 1 525
0 0 2 0 0 620
0 0 3 0 1 524
0 0 4 0 0 619

2) [view.txt](#) (6144 lines)

This is a reduced version of mapping.txt, with only 4 columns :

Crate; AMC; AMC_channel; view_type

0 0 0 0
0 0 1 1
0 0 2 0
0 0 3 1
0 0 4 0

The correspondence between crates, AMC and their IP addresses is provided in the next page

Correspondence table between crate number and the IP of its MCH

Crate	IP	# AMC cards
0	10.73.32.128	10
1	10.73.33.128	10
2	10.73.34.128	10
3	10.73.35.128	10
4	10.73.36.128	10
5	10.73.37.128	10
6	10.73.38.128	10
7	10.73.39.128	8
8	10.73.40.128	8
9	10.73.41.128	10

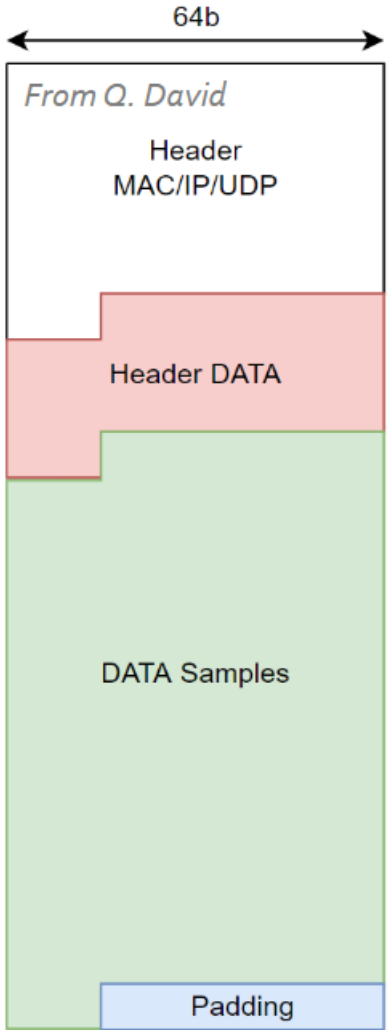
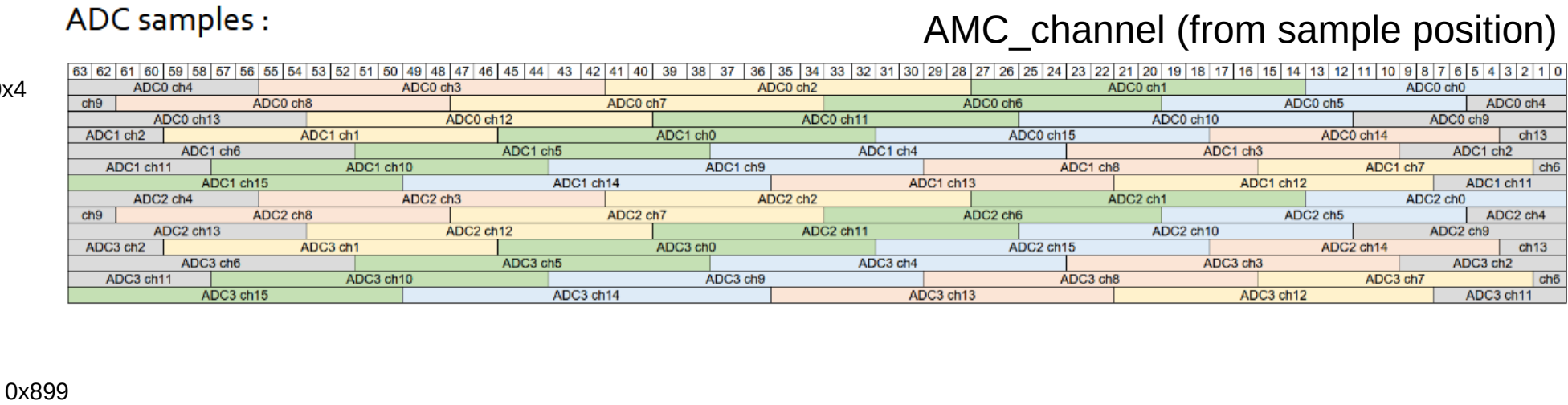
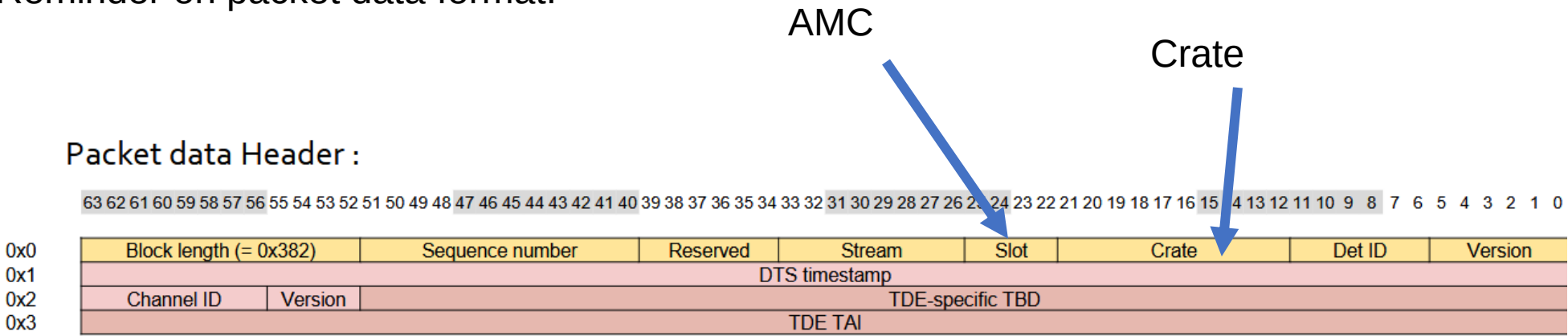
Correspondence table between AMC cards in a crate and their IPs: 1Gb network (controls), 10Gb network (data)

AMC	IP (1Gb)	IP (10Gb)
0	10.73.n.1	10.73.n.13
1	10.73.n.2	10.73.n.14
2	10.73.n.3	10.73.n.15
3	10.73.n.4	10.73.n.16
4	10.73.n.5	10.73.n.17
5	10.73.n.6	10.73.n.18
6	10.73.n.7	10.73.n.19
7	10.73.n.8	10.73.n.20
8	10.73.n.9	10.73.n.21
9	10.73.n.10	10.73.n.22

Given the particular chimney arrangement and cabling topology of NP02 a crate may read channels from different CRPs, and different views.

Crate	IP	CRP0 ind1	CRP0 ind2	CRP0 coll	CRP1 ind1	CRP1 ind2	CRP1 coll
0	10.73.32.128	x	x	x			
1	10.73.33.128	x	x	x			
2	10.73.34.128	x	x				
3	10.73.35.128				x	x	
4	10.73.36.128				x	x	x
5	10.73.37.128				x	x	x
6	10.73.38.128		x	x	x		x
7	10.73.39.128	x	x	x	x	x	x
8	10.73.40.128	x	x	x	x	x	x
9	10.73.41.128	x		x		x	x

Reminder on packet data format:



Data ADC samples, 64 consecutive ADC samples for 64 channels, 14 bit/sample, little endian :

ADC_Sample 0 for AMC_channels 0-63, ..., ADC_Sample n for AMC_channels 0-63, ... ADC_Sample 63 for AMC_channels 0-63

→ 42 B network header + 32 B data header + 64x64x14= 7168 B + 6 B padding (0x99AABBCCDDEEFF) to close the last 64 bits word boundary wrt network header = 7248 B = 906 64 bit words